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#include <SPI.h>
#include <MFRC522.h>
#include <Servo.h>
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
#include <Keypad.h>

#define SS_PIN 53
#define RST_PIN 5

MFRC522 rfid(SS_PIN, RST_PIN);

Servo lockServo;
LiquidCrystal_I2C lcd(0x27, 16, 2);

char password[] = "1234";
char enteredPassword[5];
byte index = 0;

// RFID UID
String validUID = "BE54F63";

const byte ROWS = 4;
const byte COLS = 4;

char keys[ROWS][COLS] = {
  {'1','2','3','A'},
  {'4','5','6','B'},
  {'7','8','9','C'},
  {'*','0','#','D'}
};

byte rowPins[ROWS] = {22,23,24,25};
byte colPins[COLS] = {26,27,28,29};

Keypad keypad = Keypad(makeKeymap(keys), rowPins, colPins, ROWS, COLS);

void unlockDoor()
{
  lcd.clear();
  lcd.setCursor(0,0);
  lcd.print("Access Granted");

  lockServo.write(90);      // Unlock

  delay(5000);

  lockServo.write(0);       // Lock

  lcd.clear();

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    lcd.print("Door Locked");
    delay(2000);
}

void setup()
{
    Serial.begin(9600);

    SPI.begin();
    rfid.PCD_Init();

    lockServo.attach(8);
    lockServo.write(0);

    lcd.init();
    lcd.backlight();

    lcd.setCursor(0,0);
    lcd.print("Smart Door Lock");
    delay(2000);

    lcd.clear();
    lcd.print("Scan RFID");
}

void loop()
{
    // RFID Authentication

    if (rfid.PICC_IsNewCardPresent() &&
        rfid.PICC_ReadCardSerial())
    {
        String content = "";

        for (byte i = 0; i < rfid.uid.size; i++)
        {
            content += String(rfid.uid.uidByte[i], HEX);
        }

        content.toUpperCase();

        Serial.println(content);

        if(content == validUID)
        {
            unlockDoor();
        }
        else
        {
            lcd.clear();

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        lcd.print("Invalid RFID");
        delay(2000);
    }

    rfid.PICC_HaltA();
}

// Keypad Authentication

char key = keypad.getKey();

if(key)
{
    lcd.setCursor(index,1);
    lcd.print("*");

    enteredPassword[index] = key;
    index++;

    if(index == 4)
    {
        enteredPassword[4] = '\0';

        if(strcmp(enteredPassword,password)==0)
        {
            unlockDoor();
        }
        else
        {
            lcd.clear();
            lcd.print("Wrong PIN");
            delay(2000);
        }

        index = 0;

        lcd.clear();
        lcd.print("Scan RFID");
    }
}
}

```